

GPS-based curve estimation for an adaptive pure pursuit algorithm

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Abstract. Different algorithms for path tracking have been described and implemented for autonomous vehicles. Traditional geometric algorithms like Pure Pursuit use position information to compute vehicle's steering angle to follow a predefined path. The main issue of these algorithms resides in cutting corners since no curvature information is taken into account. In order to overcome this problem, we present a sub-system for path tracking where an algorithm that analyzes GPS information off-line classifies high curvature segments and estimates the ideal speed for each one. Additionally since the evaluation of our sub-system is performed through a simulation of an adaptive Pure Pursuit algorithm, we propose a method to estimate dynamically its look-ahead distance based on the vehicle speed and lateral error. As it will be shown through experimental results, our sub-system introduces improvements in comfort and safety due to the extracted geometry information and speed control, stabilizing the vehicle and minimizing the lateral error.

Keywords: Autonomous vehicle, path tracking, look-ahead distance, lateral error, curvature estimation, pure pursuit.

1 Introduction

Autonomous vehicle navigation has become a challenged field of study during the last three decades. One of the main tasks consists in trajectory tracking. The goal of path tracking is to minimize the lateral distance between the vehicle and defined path. There are different methods to perform this task which can be classified into geometric, kinematic and dynamic models [22].

Kinematic models [7, 16] neglect vehicle dynamics as well as geometric ones, but its performance is worse than simple geometric models. Dynamic models [15], on the other hand, take into account vehicle dynamics impacting tracking performance when speed increases or curvature varies. The problem with these kind of models is their complexity, since most of the time, they are non-linear and computationally expensive. Geometric models [2, 11, 20, 21] are one of the most popular due to their good accuracy and easy implementation.

Geometric models, as their name suggest, are based on geometric relationships between the vehicle and the path. They make use of a look-ahead distance

to measure error ahead of the vehicle and can extend from simple circular arc calculations to more complicated ones involving screw theory [24]. Pure pursuit, vector pursuit and Stanley methods belong to these models. A comparison between geometric models is performed in [22] leading us to the conclusion of using Pure Pursuit as our path tracking algorithm.

The original pure pursuit method is a pure path tracking algorithm which does not consider vehicle speed to define the distance for the target point [2]. This definition results in two kind of problems: (1) when the distance is too small, it causes the vehicle to return to the path more aggressively (oscillations), and (2) when it is too big, curvature information is neglected (cutting corners). In order to calculate a convenient look-ahead distance, different factors need to be considered. One of them is speed, which is the main parameter when no visual information is used [15, 20]. On the contrary, there are studies which tend to understand how humans define their fixation points (look-ahead fixations) to drive safely [12, 13] based on time, horizon and speed. The work presented here focuses only on GPS information, and the look-ahead distance definition takes into account current vehicle speed, path geometry (curvature information) and feedback error (lateral error) of the path tracker.

This paper proposes a curve safety sub-system for path tracking (see figure 1) based on the Pure Pursuit algorithm and a dynamic look-ahead distance definition based on vehicle current speed and lateral error. The aim of this sub-system is to analyze the geometric information of the planned path and to define the convenient speed and look-ahead distance that the vehicle needs to perform more stable path tracking and reduce the lateral error. Our solution pre-processes GPS data through Ramer-Douglas-Peucker algorithm [3] to reduce the number of points. The GPS points are then analyzed to define segments which belong to a curve or a tangent line. For each curve segment, the radius, angle, length and convenient driving speed are computed. The idea behind the curve estimation relies on overcoming the cutting corners problem of the pure pursuit tracking algorithm. In other words, if a segment is detected as part of a curve, the vehicle will need to decrease speed and consider a shorter look-ahead distance; otherwise it will increase speed until reaching the maximum one and consequently considering a long look-ahead distance. Even though curvature estimation performs good, right threshold value needs to be tuned.

The rest of the paper is organized as follows. Section 2 presents a review of similar works. Section 3 introduces the basics of the Pure pursuit algorithm. Section 4 explains how curvature information is extracted with our sub-system. This information will be considered in Section 5 for velocity planning module and definition of look-ahead distance. In Section 6, analysis and discussion of the results with and without the improvements of our sub-system are presented. Finally, Section 7 concludes our work and introduces future ideas to implement.

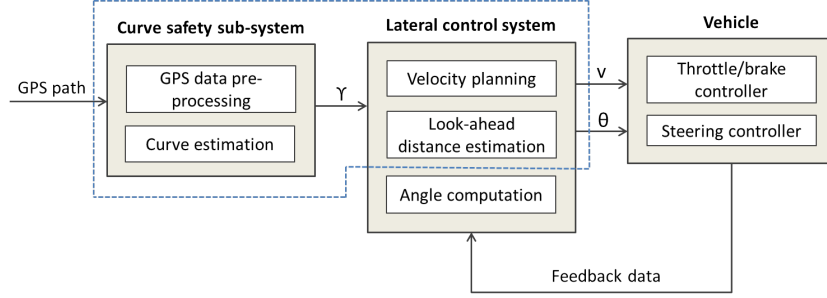


Fig. 1. Overall work-flow of the vehicle path tracking system. Blue dotted lines highlight our contributions.

2 Related works

The main related issues taken into account in our sub-system can be classified into 2 parts: (1) Curve estimation, and (2) Pure pursuit control modification. Important state of the art studies in these fields are outlined below.

2.1 Curve estimation

Road shape estimation has been considered as an essential function for road following and other applications in Advanced Driving Assistance System (ADAS).

Several studies have tried to estimate road geometry. Some of them are based on the motion analysis of other vehicles when poor visibility is present (weather changes, road hills) [4, 17, 18]. The idea behind this comes from assuming that observed leading vehicles are likely to keep following their lane [4]. Mostly, these works use different sensors like video cameras, infrared and radar sensors, as well as steering wheel and wheel speed sensors.

Some other works proposed to extract curvature information from GPS data and GIS maps with the aim to calculate the road ahead of the vehicle through a flexible road model [23]. In other cases, they are used to identify critical curves where accidents occur frequently [5, 8].

Also, researchers have focused on methods to identify curves and measure safety-related parameters using again GPS data, and GIS information. These with the objective to estimate curve radius, length and angle through the identification of straight line, clothoid and circular curves [8, 14].

There are a lot of methods to identify road geometry, nevertheless, only a few of them have focused on considering convenient velocities to perform safe trajectory with actual tracking algorithms for Unmanned Ground Vehicles (UGVs) [19–21]. Our work is based on curvature estimation [14] considering only off-line GPS information to detect curves and estimate reference velocities of each segment of the path. This estimation is calculated by considering various properties such as curvature, flatness and side friction coefficient [21].

2.2 Pure pursuit

Once curved segments are identified in the predefined path, an evaluation of the safe trajectory needs to be performed through a path tracking algorithm. Pure pursuit was chosen to perform this task in order to deal with the problem of cutting corners.

A number of works have been developed to improve the accuracy of the pure pursuit algorithm [11, 15, 20, 21]. Some of them combining a PID or PD controller together with Pure Pursuit [20, 21] in order to decrease the tracking error. Others modifying the look-ahead distance [11, 15, 20, 21], since it is the most important parameter of this algorithm.

The idea of a dynamic look-ahead distance started with [11] in which they sum up every second the distance between vehicle and path (lateral error) and the fixed look-head distance. In the original Pure Pursuit method, the definition of the look-ahead distance was a fixed term [2].

Some other authors have defined the look-ahead distance based on a certain weight and speed limits [15, 20, 21]. The magnitude of the look-ahead distance is dynamically adjusted considering min and max values for certain speed thresholds, and a weight value for a speed range. The minimum distance is defined according to what the vehicle needs to respond safely when there is an emergency stop [15]. The weight applied to certain vehicle speeds is an adjustable parameter. Usually, this parameter is fixed as presented in [15, 20, 21], but the aim of our work is to make it dynamic according to lateral error and vehicle speed.

An adaptive Pure Pursuit algorithm will be considered for our test paths modifying the distance to the goal point (target point) and taking into account the curvature information. Through simulations we will compare the results with DGPS (Differential Global Positioning System) data.

3 Path tracking algorithm

The geometric-based pure pursuit method consists in defining a goal point in the path at a certain distance ahead of the vehicle actual position, and reaching it drawing a circular arc. This algorithm iterates continuously with the goal point sliding along the path, forming a smooth tracking trajectory, as shown in figure 2 (left). It operates in a similar way humans drive, in which we fix a point at some distance ahead on the road and attempt to follow it.

The goal point is determined by the look-ahead distance (l) which is the only parameter the pure pursuit depends on. In figure 2 (G_x, G_y) represent the goal point in xy coordinates, (P_x, P_y) the projected point, R the radius, θ_{err} the heading angle needed it to reach the point.

Estimating circular arc for Pure pursuit algorithm depends only on the look-ahead distance, as seen in equation 1. For this reason, this algorithm is easy to implement. Even though it is easy, tuning the look-ahead distance could be a bit tricky since it could lead to problems defining it too small or too big [6] as mentioned previously.

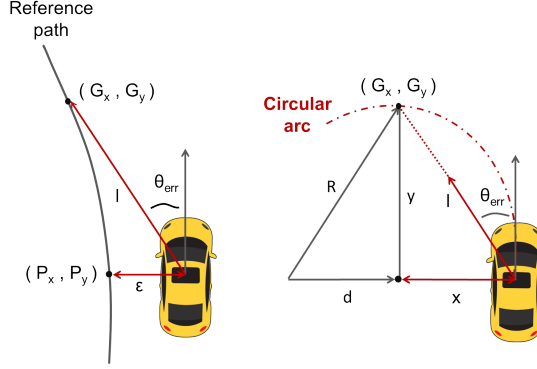


Fig. 2. Pure pursuit geometry diagram showing the path to be tracked(left) and the calculated steering curvature(right).

$$\gamma = \frac{2\theta_{err}}{l} \quad (1)$$

4 Curve safety sub-system

Detection of curves along a recorded GPS path for an unnamed ground vehicle (UGV) will help path tracking algorithms to perform a safer and more accurate route.

In order to start analyzing the path, GPS data needs to be preprocessed before to detect the points which belong to a curvature. Once the points are detected, an estimation of the radius, angle and length will be performed for each horizontal curve based on the work of Li et al. [14] with minor modifications.

4.1 GPS pre-processing

The accuracy of GPS data (15 meters) can vary depending on different factors. Among them are the quality of the GPS receiver, the environmental surroundings (tall buildings, trees, tunnels, valleys, etc.) and even the weather itself.

Differential Global Positioning System (DGPS) was used for data collection due to the improved accuracy it provides (about 10 cm). Even though the recorded data is supposed to be trust-able, there is a lot of noise which needs to be considered before trying to detect curves. In our case noisy points were removed manually in order to have a reliable route. At the same time, the amount of information to be processed for curve analysis is huge and in consequence the performance of the method will be time consuming and storage capacity dependent.

Since a recorded path is described as a sequence of points, it is possible to reduce its number considering line simplification algorithms. Douglas-Peucker

(DP) algorithm is among the most accurate line simplification algorithms with minimum shape distortions according to a comparison made with different algorithms in Ivanov et al. work [9].

DG algorithm [3] finds a similar curve to the original with less number of points. The algorithm is based on a threshold value which determines the distance at which the point should be kept or discarded for the new simplified curve. It divides recursively the line according to the following manner:

1. At the beginning, the first and last points are kept.
2. Find the point located farther away from the line segment drawn between these two points.
3. Evaluate the furthest point (best point) with the threshold (ε). If the distance between the farthest point and line segment is greater than ε , the point must be kept or discarded otherwise.
4. Recursively evaluate the points between the first and best points and between the best and end points repeating steps 2 and 3.

The drawbacks of this algorithm are its complexity and the definition of its threshold value. Its average complexity is $O(n^2 \log m)$, and in the best case $O(n \log n \log m)$, where n is the total number of points and m is the number of points retained [9]. The threshold ε is user defined. The number of output points will vary depending on its value. For this reason, the user needs to tune this value according to the desired number of points. In our case, a value of 0.05 reduced the number of points around 8.5 times and kept the information needed to evaluate curvature information.

4.2 Curve estimation

PC and PT identification method. Horizontal curve identification is represented by its point of curvature (PC), point of tangency (PT), point of intersection (PI), curve length (L), radius (R) and central angle (θ). An illustration of these curve parameters is shown in figure 3a.

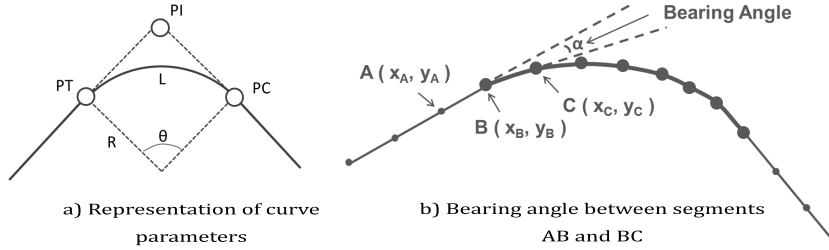


Fig. 3. Curve parameters representation

Identifying the start and end points (PT and PC) is critical for curve classification. We identify and classify the curve segments according to the method

proposed by Li et al. [14], where a large central angle describes sharp curves (dangerous curves) and large radius describes less severe curves.

The critical step for this method is how to choose the ideal threshold value (bearing angle) to identify curves between two consecutive segments. An illustration of bearing angle is presented in figure 3b.

A, B and C are consecutive points. The computation of this angle between 2 consecutive segments is performed following the formula:

$$\alpha = \cos^{-1} \left(\frac{(x_B - x_A)(x_C - x_B) + (y_B - y_A)(y_C - y_B)}{\sqrt{(x_B - x_A)^2 + (y_B - y_A)^2} \times \sqrt{(x_C - x_B)^2 + (y_C - y_B)^2}} \right) \times \frac{180}{\pi} \quad (2)$$

If the angle is greater than the threshold value, the center point is considered as part of a curve. Tuning this threshold value is very important since for long and smooth curves the bearing angle could be very small (less than 1°) and will tend to confuse the curved segments with tangent ones. Different threshold values were evaluated ending up following the optimal one as proposed in [14] (1.25°).

Secondly, and different from Li et al. work, where a compound curve (multiple curves) is considered as long as the distance between curves is less than 183 m; we defined the minimum number of consecutive points equal to 3, in order to have a candidate curve and we do not considered length between them. An evaluation with their criteria was performed and our whole path was considered a compound curve. The reasons behind this behavior could be due to: (1) their definition was given for roadway maps and in our datasets, only urban environments were considered; and (2) our datasets are still not completely free from noisy information.

Curve parameters estimation. The geometric information of each curve will be computed once the initial and ending points are identified (PC and PT). These information includes the radius (R), length (L) and central angle(θ). In order to perform these computations, the center point of the curve (O) needs to be found. These parameters can be computed following the work [14].

Besides Li et al. classification, we went further identifying sharp curves. Sharp curves deserve special attention due to the fact that it's a location where the vehicle needs to decrease speed in order to take the curvature as if a human was driving, keeping the lateral error as smaller as possible. The aim of identifying these kind of curves is related to safety issues, which is a very important point for intelligent vehicle systems [10].

Curves are classified as sharp if the radius is small or the curve angle is big. An angle range from 30° - 180° defines a dangerous curve with radius from 5-18 meters according to American Association of State Highway and Transportation Officials' (AASHTO's) "A Policy on Geometric Design of Highways and Streets" [1]. In our datasets, any curve with angle bigger than 30° is considered a sharp curve.

5 Lateral control system

Lateral control system is a path tracking system based on the pure pursuit algorithm [2]. We divided it into three units: velocity planning, look-ahead distance estimation, and steering angle computation. Our work covers only the first two units, since the aim is to show how tracking accuracy is improved taking into account path geometry.

The velocity planning component estimates the ideal speed of the vehicle taking into account the curvature information obtained previously in section 4. The look-ahead distance unit computes the distance considered to define the goal point depending on the vehicle speed and lateral error. A more detailed description of these units is explained in the following section.

5.1 Velocity planning

Velocity planning module calculates the ideal speed needed when the vehicle moves around a horizontal curve. When the vehicle drives in a curve, the force that makes it follow the curve is called centripetal force. This force does not depend on the gravity but instead on the friction between tires and the roadway. Opposite to this force, centrifugal one tries to move the vehicle outwards the curve. Centrifugal force can be reduced inclining the road into the center of the circle modeling the curve. This inclination is also known as banked angle (θ) or super-elevation angle (e). In other words, the ideal speed will be computed depending on the curvature of the path, friction coefficient and banked angle as shown in figure 4. N is the normal force, f is the friction force, μ is the friction coefficient, m is the vehicle mass, g is the gravity and F_{net} is the net force.

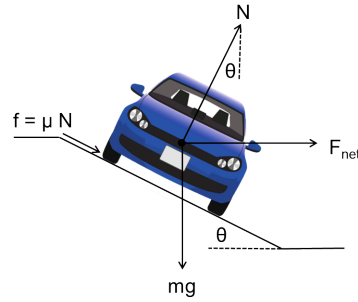


Fig. 4. Centripetal force diagram

The estimation of the ideal speed is given by equation 3 where k represents the curvature information which is the inverse of the curve radius $R = \frac{1}{k}$.

$$v = \sqrt{\frac{(e + \mu)g}{k}} \quad (3)$$

Super-elevation is defined to be between 6%-12% for rural and urban roads according to American Association of State Highway and Transportation Officials' (AASHTO's) "A Policy on Geometric Design of Highways and Streets" [1].

The side friction factor depends on the road surface conditions and vehicle speed. It is usually limited from 0.1 to 0.16 [21]. We compute the ideal speed for all the curves identified in section 4 and limit the max velocity for the path tracking algorithm to 36 kmph. Only sharp curves ended up with a speed smaller than the maximum one.

5.2 Look-ahead distance definition

The look-ahead distance value determines the accuracy of the pure pursuit algorithm. In other words, if the look-ahead distance is too short for the current speed, vehicle will be unstable, ending up in a lot of oscillations. On the other hand, if it is too long, the tracking performance will be deteriorated, resulting in the well-known problem of 'cutting corners'.

In this paper, we defined a dynamic look-ahead distance based on the vehicle speed and the lateral error as:

$$L = \begin{cases} L_{min} + v, & \text{if } T \leq 1 \\ L_{min} + Tv, & \text{otherwise} \end{cases} \quad (4)$$

where L_{min} is the minimum look-ahead distance, v is the current vehicle speed and T is a gain value. The gain value is a dynamic parameter defined according to the lateral error and the maximum speed ($T = \frac{L}{v_{max}}$). The maximum speed is 36 kmph.

The idea behind the gain T is to stabilize the vehicle faster when the distance from the vehicle to the path is big, and in consequence, removing the lateral error faster. This improved look-ahead distance definition is useful only if the tracking precision is low, otherwise there won't be much difference. A similar idea was presented in [15] but without a dynamic gain value. Most of the state of the art works which consider a dynamic look-ahead distance definition use a fixed gain value depending on speed thresholds [15, 20, 21]. The main drawback of these works is related to their accuracy because as soon as the vehicle speed increases, the lateral error will also increase.

6 Experimental results

In this section we present simulation results performed on real datasets collected with the SetCar (see fig 5) of the IRTES-SeT laboratory (Belfort, France). This vehicle is based on an electric GEM car equipped with several sensors, among them a GPS receiver and a RTK-GPS receiver, together with an embedded hard disk and a computer system (PC) to log the acquired data.

Two datasets were considered containing GPS data with different lengths and curvature information. A small description of the datasets is listed below.



Fig. 5. Equipped SeTCar

- **Dataset UTBM-1.** This path contains 7,803 original points (pts) and 882 pts after pre-processing. The vehicle was driven in urban environments through a distance of about 3.8 km. A number of 40 curves were detected following Liu et al. classification [15] and 15 sharp curves (dangerous curves) came out after our analysis. For the final evaluation, our proposed curve safety sub-system (section 4) passes this information to the lateral control system to calculate the ideal driving speed for each one.
- **Dataset UTBM-2.** The path contains 4,963 original GPS pts and only 540 after pre-processing. The distance of this track is about 2.2 km. Following Liu et al. classification [15], 22 curves were detected and only 6 sharp curves resulted after our classification. The ideal driving speed for each curve was computed once in the lateral control system and applied for the simulations.

Path tracking simulations were performed with and without considering curvature information for both datasets. When considering curves, the lateral control system knows exactly where the curves are located and estimates the necessary distance to start decelerating (2 m/s^2) before arriving to them, until reaching the desired speed at the beginning and throughout the curves. In these simulations, a maximum speed of 36kph was used.

Figures 6 and 7 show the results of UTBM-1 and UTBM-2 datasets comparing the obtained simulations against the original GPS information. Table 1 shows the complete results regarding lateral errors obtained in each curve.

The performance of our curve safety sub-system showed significant improvements for most of the curves, but similar or inclusive less accuracy in some others. At the same time it is remarkably to note that standard deviation (SD) decrease by half in each path (see table 1) which makes our proposed subsystem more reliable.

For UTBM-1 dataset, tracking error is reduced by about 25% and around 35% for UTBM-2 dataset. While accuracy is improved for most of the curves in last dataset (see Table 1), error in curve 4 increased around 34%. The same behavior occurred in curves 5, 8, 11 and 14 of UTBM-1 dataset (see Table 1). The reason behind this increment is due to the fact that the lateral control

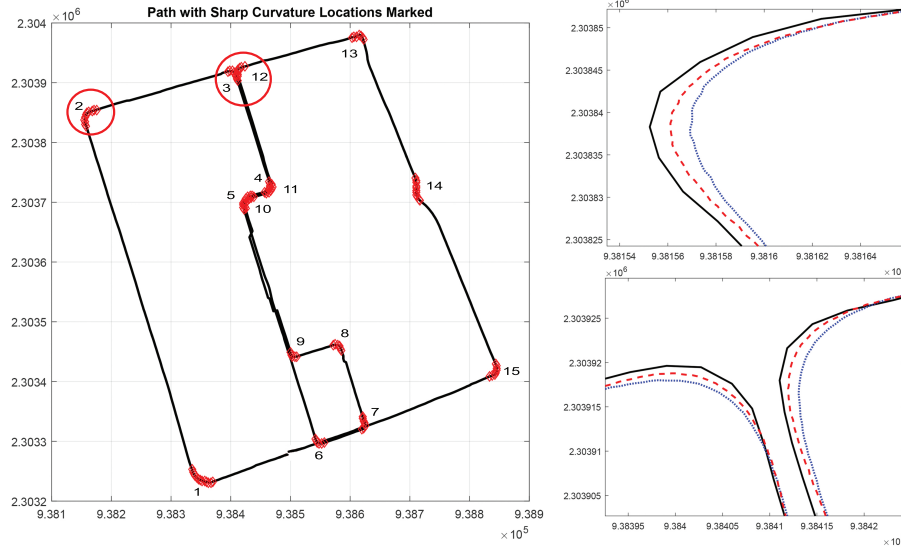


Fig. 6. Lateral errors comparison for UTBM-1 dataset. Solid black line (—) represents the original path, dotted blue line ($\cdots$) is the simulated path, and dashed red line (--) is the simulated path considering curvature information.

system (without considering curvature information) reaches smaller speeds than the ones computed as convenient for the curves. As a result of low speed, small look-ahead distance is defined, reducing tracking error but emerging oscillations. These oscillations are one of the problems of the pure pursuit algorithm and with our system we have proven, through simulations, that they can be reduced.

Another problem of the Pure Pursuit algorithm is the 'cutting corners'. A visual comparison is shown in figures 6 and 7 where the simulated path, considering curvature information, sticks closer to the original path.

In order to compare both methods, the average root mean square error (RMSE) and standard deviation (σ) per method are calculated regarding lateral errors obtained in curves for each path.

$$\overline{RMSE} = \frac{1}{2} \sum RMSE \quad (5)$$

$$\sigma = \left(\frac{1}{2} \sum (RMSE - \overline{RMSE})^2 \right)^{\frac{1}{2}} \quad (6)$$

The comparison results are shown in figure 8. The tracking errors with our subsystem improved significantly **32%** over the pure pursuit without considering curvature information. Here the standard deviation only tells us that errors depend on the characteristics of the path itself.

With the aim to prove that a wrong definition of the look-ahead distance can affect considerably the accuracy of the tracking algorithm, we compared our

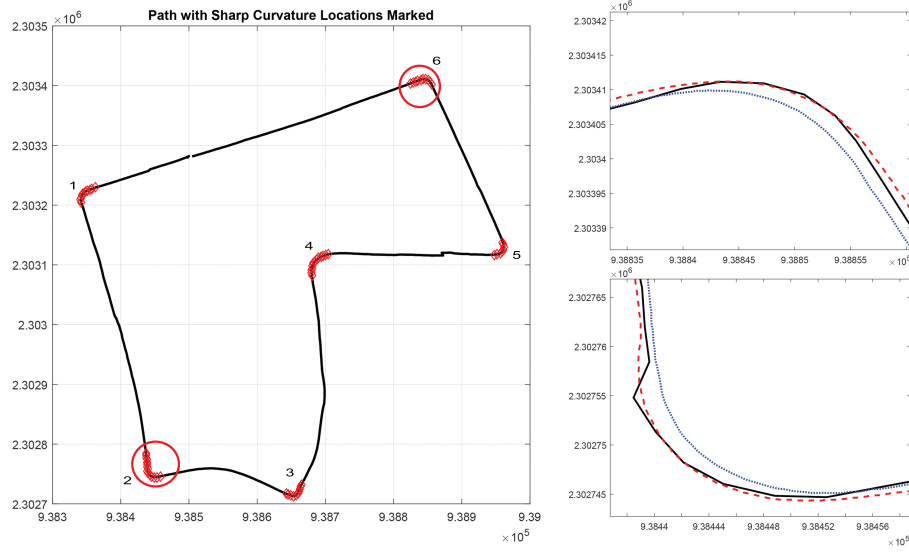


Fig. 7. Lateral errors comparison for UTBM-2 dataset. Solid black line (—) represents the original path, dotted blue line (···) the simulated path, and dashed red line (--) the simulated path considering curvature information.

definition (section 5.2) with the one of Liu et Duan [15]. Our curvature system was active for both simulations with a max speed of 36 kmph. The test was made with UTBM-2 dataset and the average lateral error resulted in 1.51 meters with Liu et Duan's definition; almost 3 times more than with our look-ahead distance definition (0.54 m).

Even though our implementation have proven to be easy, robust and reliable; some limitations still remain. The main one refers to the data preprocessing for smoothing the GPS path and getting rid of noisy information. Since the curvature identification depends on the path itself, a bad or noisy track will end up with really bad results.

Another limitation is the look-ahead distance definition. Our proposal makes the vehicle reach the path faster (but more stable) when the lateral error is big, otherwise, the look-ahead distance definition depends only on the vehicle speed. Although most of the literature share the same idea of defining the look-ahead distance based on the speed, tuning this parameter could result tricky and hence an improved definition should be considered taking into account more information.

7 Conclusions

We have proposed a look-ahead definition, convenient speed estimation and curve safety sub-system for path tracking algorithms based only on GPS information.

Table 1. Lateral errors according to Pure Pursuit simulations. Results are expressed in meters.

Curve	UTBM-1		UTBM-2	
	Original	Considering curves	Original	Considering curves
1	0.6649	0.6577	0.7446	0.4672
2	1.0043	0.6409	0.9107	0.4163
3	1.1658	0.7097	1.1627	0.5916
4	1.2618	0.6535	0.5161	0.6919
5	0.2658	0.3346	0.8626	0.4823
6	0.5092	0.4508	0.8013	0.6172
7	0.6772	0.6135		
8	0.4286	0.4393		
9	1.2744	0.7112		
10	1.1698	0.9222		
11	0.2656	0.2763		
12	1.1369	0.8036		
13	0.8694	0.5742		
14	0.5402	0.6990		
15	0.9733	0.6803		
Average	0.8138	0.6111	0.833	0.5444
SD	0.3455	0.1665	0.1936	0.0962

Our sub-system, which works off-line, is capable to reduce significantly tracking errors and solve the well known problem of cutting corners for Pure Pursuit.

The main advantage of our method is its generality, since it can be adopted for different tracking algorithms and still increase their accuracy. Since the path geometry is analyzed automatically and the ideal driving speed is computed with the output obtained; applying the desired speed at each curve results in an easy task for the tracking system. The only requirement needed is a set of GPS points free of noise to be able to identify correctly the horizontal curves.

In conclusion, our sub-system proved to be:

1. Simple and robust: it works only with GPS information and detects accurately sharp curves which are the ones that need special attention for safety and comfort issues.
2. Accurate: it reduces the tracking error significantly throughout the curves making the vehicle more stable while driving at the convenient speed and considering an optimal look-ahead distance. This speed is computed following physics laws and considering the friction coefficient and super-elevation angle for urban roads [1].
3. Safe: the aim of any intelligent vehicle is to provide safety to the user. If the accuracy of the path tracking algorithm is good enough, we assure the user that the vehicle will not deviate and will reach its destination without any problems.

At the moment, only simulations were executed, but the sub-system will be tested in real vehicle to prove its performance. An interesting direction for future

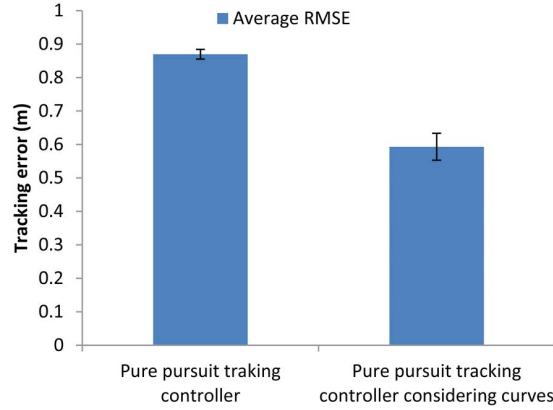


Fig. 8. Average and standard deviation of the lateral errors.

research would be to try different methods to remove GPS noise and simplify the path, like RANSAC and variations of Douglas Peucker algorithm. Also, more sensors (like cameras, lidar, or radar) need to be considered in case that unwanted moving objects appear in the predefined path. Consequently, methods to avoid obstacles and plan new trajectory will be necessary.

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References

1. Book, G.: A policy on geometric design of highways and streets. AASHTO, 5th Edition, ISBN 49, 1-56051 (2004)
2. Coulter, R.C.: Implementation of the pure pursuit path tracking algorithm. Tech. rep., DTIC Document (1992)
3. Douglas, D.H., Peucker, T.K.: Algorithms for the reduction of the number of points required to represent a digitized line or its caricature. *Cartographica: The International Journal for Geographic Information and Geovisualization* 10(2), 112-122 (1973)
4. Eidehall, A., Pohl, J., Gustafsson, F.: Joint road geometry estimation and vehicle tracking. *Control Engineering Practice* 15(12), 1484-1494 (2007)
5. Fitzsimmons, E.J., Lindheimer, T.E., Schrock, S.D., Gonterwitz, K.: Evaluation of large truck crashes at horizontal curves on two-lane rural highways in kansas. In: *Transportation Research Board 92nd Annual Meeting*. No. 13-2340 (2013)
6. Giesbrecht, J., Mackay, D., Collier, J., Verret, S.: Path tracking for unmanned ground vehicle navigation. DRDC Suffield TM 224 (2005)

7. Girbés, V., Armesto, L., Tornero, J., Solanes, J.E.: Continuous-curvature kinematic control for path following problems. In: 2011 IEEE/RSJ International Conference on Intelligent Robots and Systems. pp. 4335–4340. IEEE (2011)
8. Hans, Z., Souleyrette, R., Bogenreif, C.: Horizontal curve identification and evaluation. Tech. rep. (2012)
9. Ivanov, R.: Real-time gps track simplification algorithm for outdoor navigation of visually impaired. *Journal of Network and Computer Applications* 35(5), 1559–1567 (2012)
10. Kannan, S., Thangavelu, A., Kalivaradhan, R.: An intelligent driver assistance system (i-das) for vehicle safety modelling using ontology approach. *International Journal of Ubicomp* 1(3), 15–29 (2010)
11. Kelly, A.: A feedforward control approach to the local navigation problem for autonomous vehicles. Carnegie Mellon University, the Robotics Institute (1994)
12. Lehtonen, E., Lappi, O., Koirikivi, I., Summala, H.: Effect of driving experience on anticipatory look-ahead fixations in real curve driving. *Accident Analysis & Prevention* 70, 195–208 (2014)
13. Lehtonen, E., Lappi, O., Kotkanen, H., Summala, H.: Look-ahead fixations in curve driving. *Ergonomics* 56(1), 34–44 (2013)
14. Li, Z., Chitturi, M., Bill, A., Noyce, D.: Automated identification and extraction of horizontal curve information from geographic information system roadway maps. *Transportation Research Record: Journal of the Transportation Research Board* (2291), 80–92 (2012)
15. Liu, R., Duan, J.: A path tracking algorithm of intelligent vehicle by preview strategy. In: Control Conference (CCC), 2013 32nd Chinese. pp. 5630–5635. IEEE (2013)
16. Low, C.B., Wang, D.: Gps-based path following control for a car-like wheeled mobile robot with skidding and slipping. *IEEE Transactions on control systems technology* 16(2), 340–347 (2008)
17. Lundquist, C., Schon, T.B.: Road geometry estimation and vehicle tracking using a single track model. In: Intelligent Vehicles Symposium, 2008 IEEE. pp. 144–149. IEEE (2008)
18. Lundquist, C., Schön, T.B.: Joint ego-motion and road geometry estimation. *Information Fusion* 12(4), 253–263 (2011)
19. Park, B., Han, W.Y.: Reference velocity estimation method for on-road unmanned ground vehicle. In: Ubiquitous Robots and Ambient Intelligence (URAI), 2013 10th International Conference on. pp. 230–231. IEEE (2013)
20. Park, M.W., Lee, S.W., Han, W.Y.: Development of lateral control system for autonomous vehicle based on adaptive pure pursuit algorithm. In: Control, Automation and Systems (ICCAS), 2014 14th International Conference on. pp. 1443–1447. IEEE (2014)
21. Park, M., Lee, S., Han, W.: Development of steering control system for autonomous vehicle using geometry-based path tracking algorithm. *ETRI Journal* 37(3), 617–625 (2015)
22. Snider, J.M.: Automatic steering methods for autonomous automobile path tracking. Robotics Institute, Pittsburgh, PA, Tech. Rep. CMU-RITR-09-08 (2009)
23. Wang, C., Hu, Z., Uchimura, K.: Precise curvature estimation by cooperating with digital road map. In: Intelligent Vehicles Symposium, 2008 IEEE. pp. 859–864. IEEE (2008)
24. Wit, J.S.: Vector pursuit path tracking for autonomous ground vehicles. Tech. rep., DTIC Document (2000)